

Alireza Sani

Civil and environmental engineering, then software

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Open to work · Preferred: part-time

SUMMARY

Civil and environmental work comes first: climate observations, water systems, and structural calculations, delivered as software other people can run. Python, Android, and web services are the tools. A separate set of public repositories covers trading automation and Web3.

SELECTED WORK

ClimaScope. Live climate observatory for Android and the browser. Greenhouse gases, air quality, GloFAS river discharge, and a HighResMIP outlook. naomi197.github.io/climascope

CleanFlow Ghana. FastAPI service that records water-pollution reports and ranks them by urgency.

Water Network Optimizer. Pipe sizing and Hazen-Williams head-loss calculations for pressurized networks.

Truss Structural Optimizer. Two-dimensional truss analysis and minimum-weight section sizing in Python.

Beam Solver. Shear, moment, and deflection diagrams for a simply supported beam, with PDF reports.

Climate Assistant. Android client for current weather, a five-day forecast, and air quality from Open-Meteo.

TreeGrow. Android records of planted trees stored on the device.

Trading and Web3 automation

Published Python tools for price alerts, order-book collection, paper trading, and funding-rate scans, plus USDC invoice prototypes and a Rust grant-escrow contract. These are portfolio repositories, separate from the engineering work above.

SKILLS

Civil and environmental. Climate data, water quality, pipe networks, truss analysis, beam diagrams.

Software. Python, FastAPI, Kotlin, Jetpack Compose, TypeScript, Android, pytest, REST, Linux.

Web3. Exchange and wallet integrations, EIP-712 signature checks, async Python, WebSockets.

LANGUAGES

Persian, native. English, working.